Period of a Limit Cycle

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1 Introduction

In this homework we will numerically estimate the period of the limit cycle of the differential equation

$$x''(t) - 4\left(1 - x^2\right)x'(t) + x = 0$$

to at least 10 decimal places of accuracy.

To do so, we integrate the system with RK4, locate the period by detecting sign change ($- \to +$) of the derivative. We get the precise time of the sign change with the bisection method. We further refine the estimate by iterating and computing multiple sign changes, as they get more and more accurate as the system converges into the limit cycle.

Finally, we estimate multiple times, halving the integration step size each time until we get enough precision.

2 Mathematical Background

2.1 Reduction to a First-Order System

RK4 is formulated for first order systems, so we introduce

$$u_1 = x \qquad u_2 = x'$$

which turns the second-order ODE into the first order system

$$u_1' = u_2 \qquad u_2' = 4\left(1 - u_1^2\right)u_2 - u_1$$

2.2 Runge-Kutta 4 (RK4)

Given $u' = f(u)$ and a step size h, one RK4 step from u is

$$k_1 = f(u) \qquad k_2 = f\left(u + \tfrac{h}{2}k_1\right) \qquad k_3 = f\left(u + \tfrac{h}{2}k_2\right) \qquad k_4 = f\left(u + hk_3\right)$$

$$u_{\text{next}} = u + h\left(\frac{1}{6}k_1 + \frac{1}{3}k_2 + \frac{1}{3}k_3 + \frac{1}{6}k_4\right)$$

2.3 Detecting the Period

On the limit cycle, the velocity $u_2 = x'$ changes sign from $- \to +$ exactly once per period. We therefore integrate and whenever we detect $u_2 < 0$ at one step and $u_2 > 0$ at the next, we know a crossing occurred in that step interval. The exact crossing time is then computed with bisection method. The period is then estimated as the time between two consecutive crossings.

3 Result

Starting from an initial $u_0 = (1, 0)$ with initial step size $h_0 = 10^{-3}$, we got period estimate of

$$\boxed{T \approx 10.203523691042903}$$

More than 10 decimal digits of accuracy.
The estimation was obtained with step size $h = 2.5 \times 10^{-4}$.

4 Plots

In figure 1 we can see the trajectory of (x, x') starting from $(1, 0)$.
The trajectory spirals outward from the initial condition and settles onto the closed limit cycle loop.

In figure 2 shows $x(t)$ once the trajectory has settled to the limit cycle.

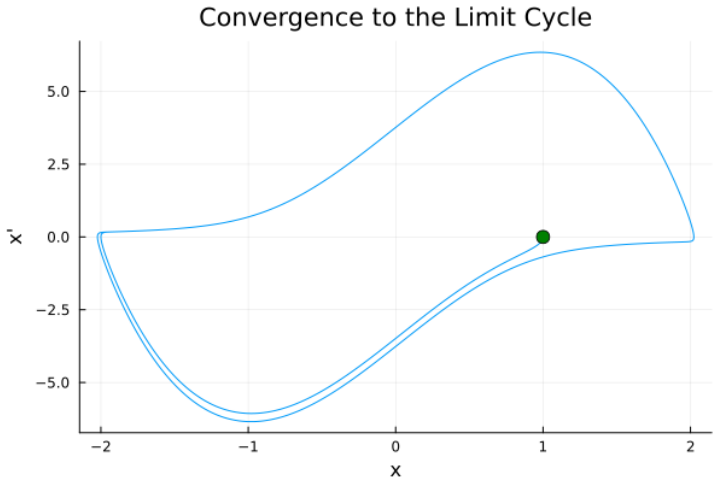


Figure 1: Trajectory of (x, x') starting from $(1, 0)$ converging onto the limit cycle

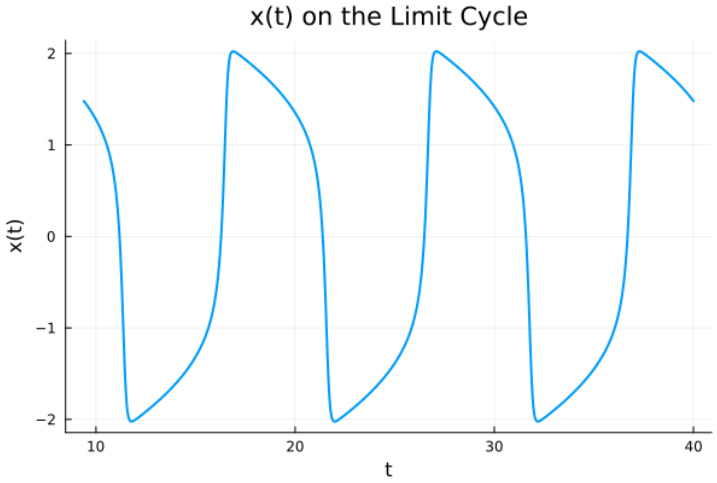


Figure 2: $x(t)$ over three periods once the trajectory has converged to the limit cycle